Materials Characterization 131 (2017) 285–297

Contents lists available at ScienceDirect

Materials Characterization

journal homepage: www.elsevier.com/locate/matchar

Phase transformation and microstructure evolution of an ultra-high strength Al-Zn-Mg-Cu alloy during homogenization

Da Xu[a], Zhihui Li[a,*], Guojun Wang[b], Xiwu Li[a], Xinyu Lv[b], Yong'an Zhang[a], Yunqiang Fan[a,b], Baiqing Xiong[a]

[a] State Key Laboratory of Nonferrous Metals and Processes, General Research Institute for Nonferrous Metals, No.2 Xinjiekou Wai Street, Beijing 100088, China
[b] Northeast Light Alloy Co., Ltd., Harbin 150060, China

ARTICLE INFO

Keywords:
Al-Zn-Mg-Cu alloy
Homogenization
Phase transformation
Microstructure
Diffusion

ABSTRACT

Homogenization treatment of an ultra-high strength Al-Zn-Mg-Cu alloy was studied by SEM, TEM, DSC, hardness and electrical conductivity test. The second phases in as-cast 7A56 alloy are AlZnMgCu, Al_2Cu and Al_7Cu_2Fe. AlZnMgCu phase has a similar lattice structure of $MgZn_2$. After 380 °C homogenization treatment, transformation from AlZnMgCu to Al_2CuMg occurred; after 470 °C homogenization treatment, AlZnMgCu were mainly directly dissolved into the matrix with no Al_2CuMg discovered. That phenomenon was caused by the diffusion behavior of Zn and Cu at two different temperatures that Zn diffuses rapidly even at 380 °C while Cu shares a much lower diffusion rate. Much of the AlZnMgCu phase can be dissolved at 470 °C for a short time while at 380 °C it was seldom eliminated. After homogenized at 470 °C for 24 h, the dissolution process gradually reached into a balance with more than 95% had been eliminated and prolonging time imposes no evident influence on promoting the dissolution. Hardness and electrical conductivity are highly consistent with the vibration of AlZnMgCu phase, with the dissolution of which the hardness increases while electrical conductivity decreases.

1. Introduction

Al-Zn-Mg-Cu alloy (7xxx series) has been widely used in several manufacturing industries due to its high strength, high ductility and low density [1–5]. As a typical precipitation strengthening alloy, the mechanical properties of the Al-Zn-Mg-Cu alloy basically depend on the volume fraction and the distribution of precipitates generated during the solution and aging treatments [6–9]. As part of that practice, it is an effective method to strengthen of the alloy by improving alloying element content for enhancing the precipitation during aging treatment. Recently, several high Zn content alloys, such as 7056, 7095, have been successfully developed and applied in aircraft manufacturing.

However, the method above will result in a new problem that more coarse second phases, such as AlZnMgCu and Al_2CuMg, can be formed when the temperature rapidly falls down during the cooling process [10–14]. As a result, it will be more difficult to eliminate these phases during the heat treatments. If remained in the Al-matrix, these second phases will depress the strengthening effect because lots of alloying atoms were detained in the coarse phases and therefore will not come into being metastable $MgZn_2$ phase that plays a pivotal role in strengthening alloys. On the other hand, the remained second phases are easily to become the crack source and subsequently provide the path of crack propagation, significantly damaging the mechanical properties of the alloy [15–18]. Homogenization treatment is an indispensable process to eliminate the second phases as much as possible and lower the difficulty of subsequent solute treatment [19–23].

Homogenization treatment of Al-Zn-Mg-Cu alloy has been studied for many years [24–28]. However, most of them merely focus on exploring a proper homogenization temper or studying phase transformation at higher temperature but not discussing the diffusion behavior at a large temperate range. 7A56 alloy, recently registered in China, is a brand-new Al-Zn-Mg-Cu alloy with high Zn content for manufacturing

* Corresponding author.
E-mail address: lzh@grinm.com (Z. Li).

http://dx.doi.org/10.1016/j.matchar.2017.07.011
Received 23 March 2017; Received in revised form 6 June 2017; Accepted 2 July 2017
Available online 08 July 2017
1044-5803/ © 2017 Elsevier Inc. All rights reserved.

Table 1
Chemical compositions of 7A56 alloy (wt%).

Zn	Mg	Cu	Zr	Fe	Si	Al
9.0	2.1	2.0	0.1	0.056	< 0.01	Bal.

Table 2
Chemical compositions of second phases in Fig. 1 (at.%).

Point	Al	Zn	Mg	Cu	Fe	Phase
A	38.09	23.27	26.03	12.62	–	AlZnMgCu
B	70.14	3.57	4.85	21.45	–	Al_2Cu
C	70.73	6.35	4.58	9.43	8.92	Al_7Cu_2Fe

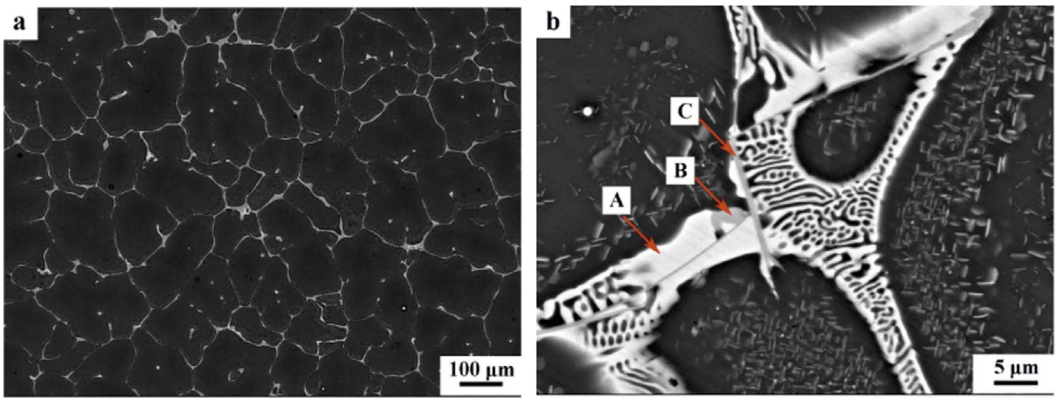

Fig. 1. SEM images of as-cast 7A56 alloy.

ultra-high strength plates. Homogenization treatment and relative discussions of this alloy has not been publicly reported yet. This article will mainly focus on two segments: 1) to discuss the phase transformation of AlZnMgCu phase at two different temperatures during homogenization; 2) to study dissolution of AlZnMgCu phase process and how it affects the electrical conductivity and hardness of the alloy.

2. Experiments

The studied 7A56 alloy was manufactured by direct-chill casting with pure Zn, pure Mg, pure Cu and Al-Zr master alloy. Chemical composition of the experienced ingot is listed in Table 1.

Samples with the size of 10 × 10 × 10 mm were directly cut from the center part of the ingot for homogenization treatment in the furnace. Samples were homogenized at 380 °C and 470 °C from 1 h to 72 h, respectively. Once taken from the furnace, samples were water quenched immediately in order to maintain the homogenized microstructures.

A JEOL JSM 7001F scanning electron microscope (SEM) attached with Energy Dispersive Spectroscopy (EDS) was used for microstructure observation. A FEI-F20 transmission electron microscope (TEM) operated with 200 kV was applied for obtaining the selected area electron diffraction (SAED) patterns of the second phases to identify and ascertain the crystal structure of the studied phases.

Thick sheets with a diameter of 3 mm were cut from the samples for Differential Scanning Calorimetry (DSC) test at a rage from 20 °C to 560 °C. The heating rate is 10 °C/min. Enthalpy of the endothermic peaks was calculated by integral of the curve.

Electrical conductivity was tested by a WD-Z eddy current conductivity meter within 5 min after water quenching. Hardness was conducted by a Rockwell Hardness Tester. Both two tests were carried out at room temperature.

3. Results and Discussion

3.1. Second Phases of As-cast 7A56 Alloy

Fig. 1 shows the SEM images of second phases in as-cast 7A56 alloy. According to the EDS analysis shown in Table 2, three kinds of second phases can be observed, including AlZnMgCu, Al_2Cu and Al_7Cu_2Fe. AlZnMgCu phase is the predominant second phase in as-cast 7A56 alloy which demonstrates a white contrast and a typical feature of eutectic phase. Selected area electron diffraction (SAED) pattern (Fig. 2a) testifies that the lattice structure of the AlZnMgCu is similar with $MgZn_2$, known as the hexagonal structure. Also, Al_2Cu and Al_7Cu_2Fe show a different contrast and shape compared with the AlZnMgCu phase, both of which are confirmed with tetragonal structure by SAED pattern (Fig. 2b and c). Given the low amount of Al_2Cu and insolubility of high melting point Al_7Cu_2Fe, what the article mainly discussed is the phase transformation and dissolution of AlZnMgCu phase ignoring the

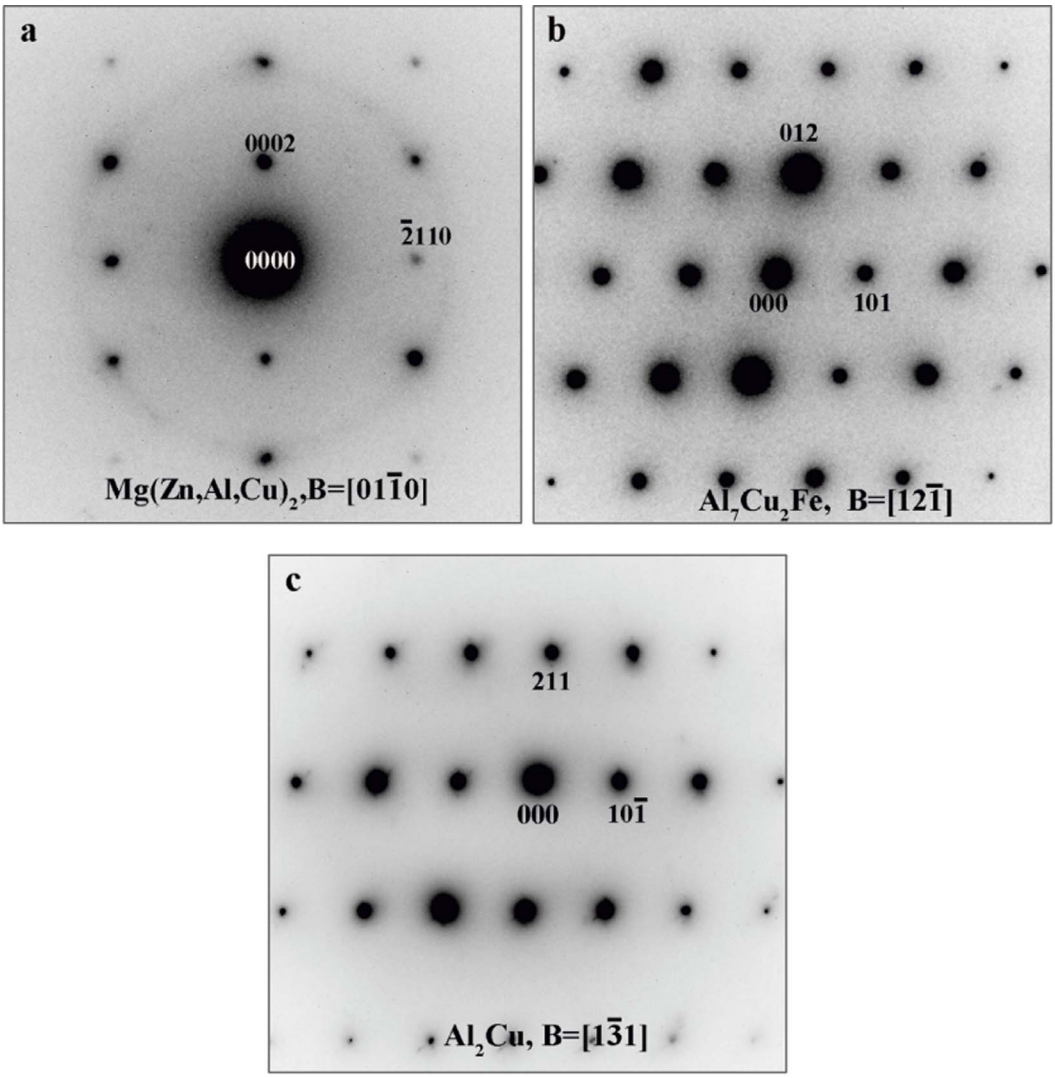

Fig. 2. SAED patterns of second phases in as-cast 7A56 alloy.
(a) $Mg(Zn,Al,Cu)_2$; (b) Al_7Cu_2Fe; (c) Al_2Cu.

influence of other two phases. Besides, with carefully observation, no Al_2CuMg phase is found in the as-cast 7A56 alloy.

3.2. Phase Transformation of AlZnMgCu Phase during Homogenization

Fig. 3 shows the SEM images of second phases in 7A56 alloy after 380 °C homogenization treatment. Table 3 shows the EDS result of the second phases in Fig. 3. Most of the AlZnMgCu phase keeps its original eutectic structure with good continuous borders even homogenized for 72 h. A new second phase with grey contrast emerged adhering to the AlZnMgCu phase. Based on the fact that the atomic ratio of Al:Cu:Mg is approximately 2:1:1 with a low content of Zn, it can be inferred that this phase is Al_2CuMg. TEM images and SAED pattern (Fig. 4) result upholds that assumption, definitely pointing out that the newly transformed Al_2CuMg is an orthogonal crystal structure. Fig. 5 demonstrates the map scanning of the area A in Fig. 4, showing a lack of Zn and an aggregation of Cu in the interior of Al_2CuMg phase (Table 4 shows the chemical compositon obtained by EDS attached to TEM). In summary, it can be concluded that the AlZnMgCu phase can transform into Al_2CuMg under 380 °C homogenization treatment.

Fig. 6 shows the SEM images of 7A56 alloy after 470 °C homogenization treatment. Table 5 shows the EDS result of second phases in Fig. 6. Only a few AlZnMgCu phase remains, the basic framework of which has been almost completely destroyed under 470 °C. AlZnMgCu

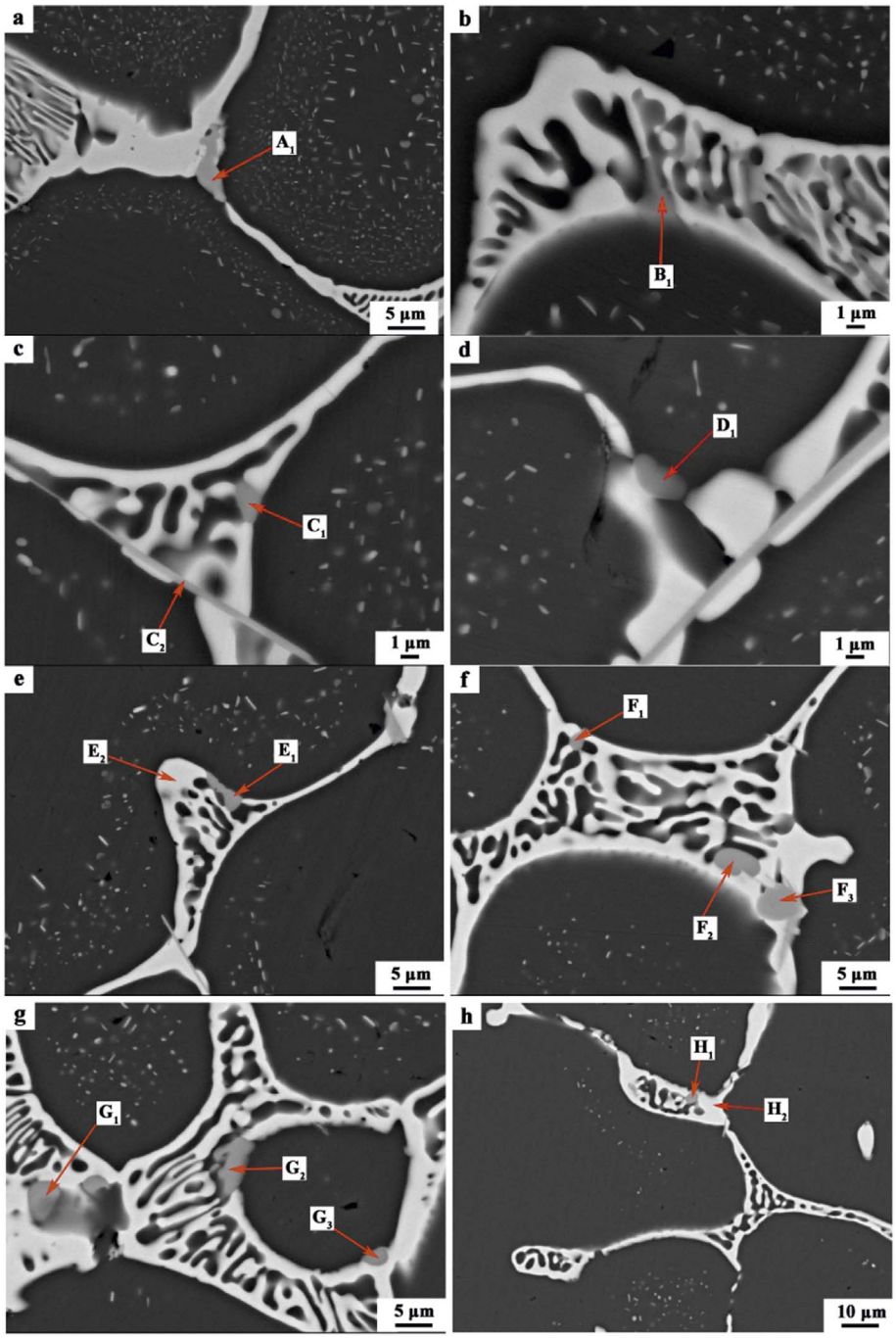

Fig. 3. SEM images of second phases in 7A56 alloy homogenized at 380 °C:
(a) 1 h; (b) 2 h; (c) 6 h; (d) 12 h; (e) 24 h; (f) 36 h; (g) 48 h; (h) 72 h.

phase had been directly dissolved into the matrix and no Al_2CuMg was found even after several repeated experiments. That is a significant difference compared with the phases after the 380 °C/24 h treatment. In addition, Al_7Cu_2Fe still remains a sharp shape with no obvious change.

Table 6 shows the solid diffusion coefficient of Cu, Mg and Zn in Al-matrix at two different temperatures. At 380 °C, D_{Cu}/D_{Zn} = 1:6.7; at 470 °C, D_{Cu}/D_{Zn} = 1:4.9. Fig. 7 shows the relationship between D_{Zn}/D_{Cu} and temperature. As the temperature increases, the value of $D_{Zn}/$

Table 3
EDS result of second phases in Fig. 3 (at.%).

Label	Al	Zn	Mg	Cu	Fe	Possible phase
A_1	67.67	2.84	–	29.49	–	Al_2Cu
B_1	68.16	5.06	13.11	13.68	–	Al_2CuMg
C_1	57.87	4.04	17.55	20.54	–	Al_2CuMg
C_2	73.40	5.80	4.09	11.83	4.87	Al_7Cu_2Fe
D_1	73.49	5.15	10.74	10.62	–	Al_2CuMg
E_1	55.85	3.38	18.26	22.51	–	Al_2CuMg
E_2	19.09	36.13	28.47	16.32	–	AlZnMgCu
F_1	57.85	6.13	17.18	18.84	–	Al_2CuMg
F_2	51.29	3.51	21.12	24.08	–	Al_2CuMg
F_3	68.77	2.30	0.73	12.78	15.42	Al_7Cu_2Fe
G_1	51.95	3.77	19.61	24.67	–	Al_2CuMg
G_2	55.33	3.17	19.78	21.71	–	Al_2CuMg
G_3	69.67	2.79	12.45	15.09	–	Al_2CuMg
H_1	54.87	2.80	21.94	20.39	–	Al_2CuMg
H_2	28.13	28.45	30.43	13.00	–	AlZnMgCu

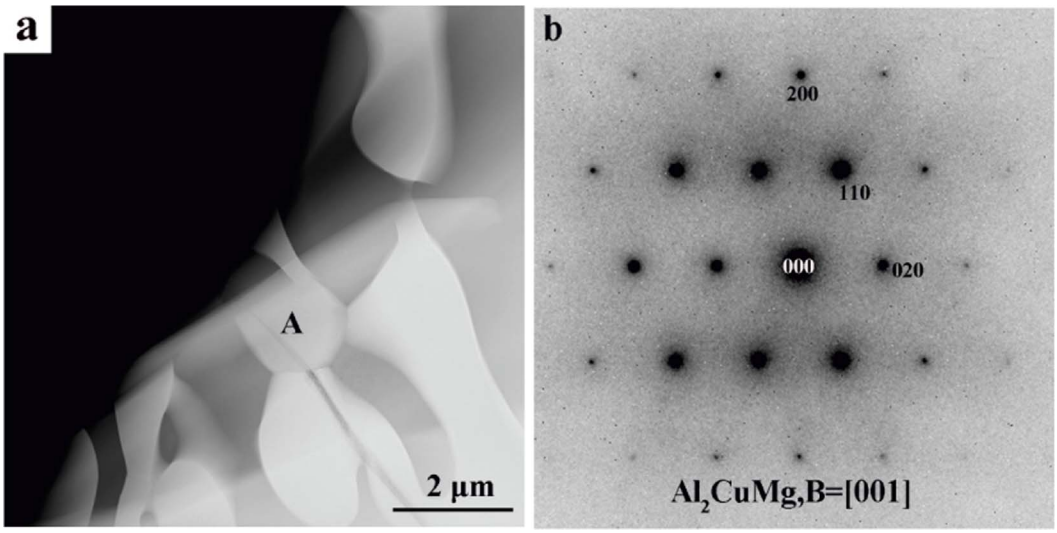

Fig. 4. Second phases in 7A56 alloy homogenized at 380 °C/24 h:
(a) TEM image; (b) SAED pattern of area A in a.

D_{Cu} slowly decreases. That means the difference of diffusion coefficient between Zn and Cu atoms decreases with the temperature increasing. At 380 °C, the difference of diffusion behavior between Zn and Cu is much bigger and Cu can easily remained in the pre-existing AlZnMgCu phase while Zn can diffuse easier than Cu. Theoretically, it can simultaneously promote the diffusion of Zn and maintain large amount of Cu atoms to form the Al_2CuMg phase at the positions where chemical composition permitted. However, at 470 °C, the diffusion speed of Cu accelerates quickly due to the high temperature. It is more difficult to sustain the two conditions above. As a result, it can hardly to transform to AlZnMgCu and but directly dissolve into matrix instead at 470 °C.

To support the view above, the content of Zn and Cu in Al-matrix during homogenization was detected by EDS (shown in Fig. 8) attached to SEM. At 380 °C, the content of Cu did not show significant vibration, indicating a quite low dissolution of Cu in practice. The content of Zn increases more under the same condition, showing a stronger tendency of diffusion which means Zn can start to diffuse at 380 °C. At 470 °C, however, both content of Zn and Cu increases significantly and demonstrates a quite diffusion behavior compared with that under 380 °C. Based on this result, the premise that Cu seldom diffuse at 380 °C seems reasonable and can be explained the phase transformation of Al_2CuMg.

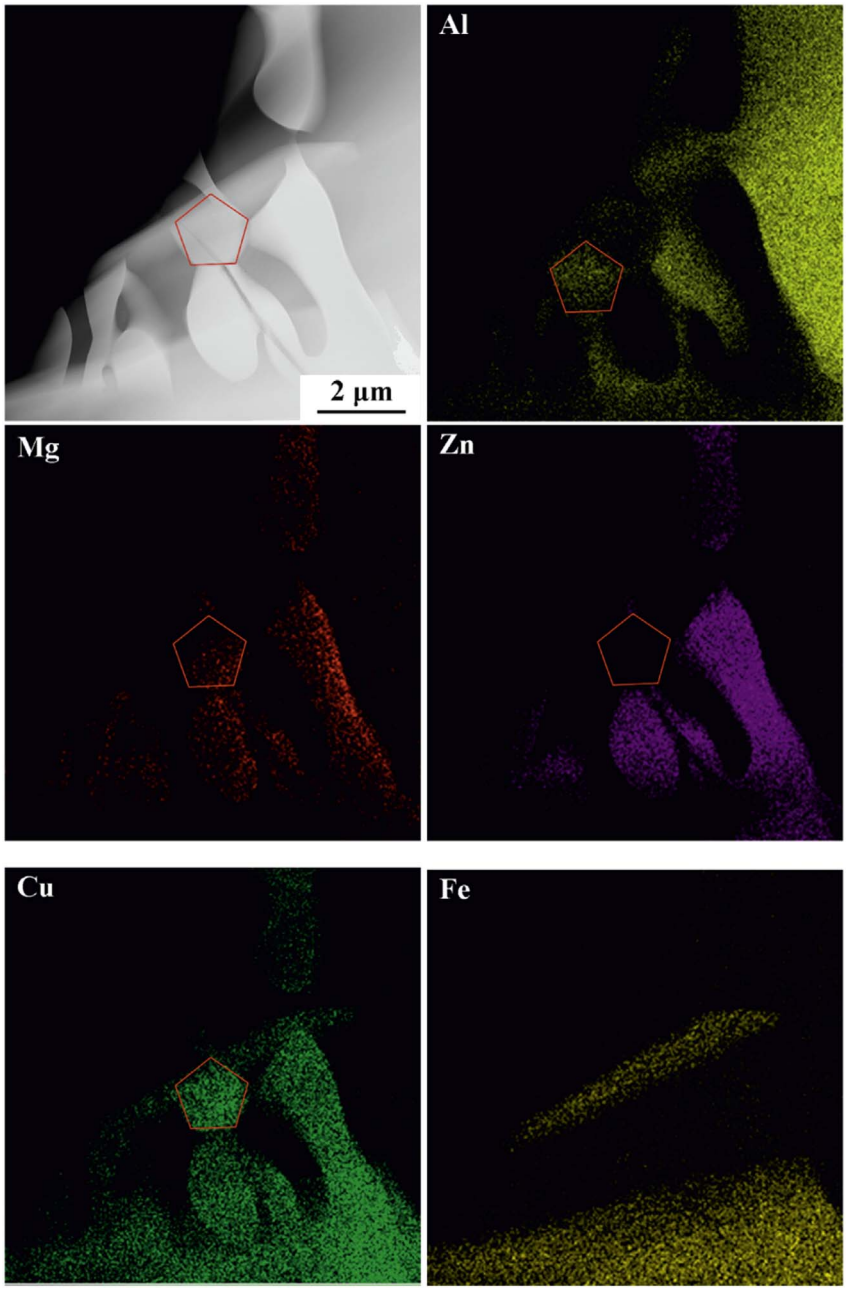

Fig. 5. Map scanning result of area A obtained by TEM.

Table 4
EDS result of area A in Fig. 4a obtained by TEM (at.%).

Al	Zn	Mg	Cu
43.00	2.26	18.95	35.79

3.3. Dissolution of AlZnMgCu Phase during Homogenization

Figs. 9 and 10 show the SEM images of 7A56 alloy after 380 °C and 470 °C homogenization treatment, respectively. At 380 °C, the AlZnMgCu phase seldom dissolved at even the time was prolonged to 72 h. There is no obvious deduction of the AlZnMgCu compared that of as-cast alloy. However, after 470 °C/1 h treatment, in a short time, AlZnMgCu phase has decreased tremendously that the border became

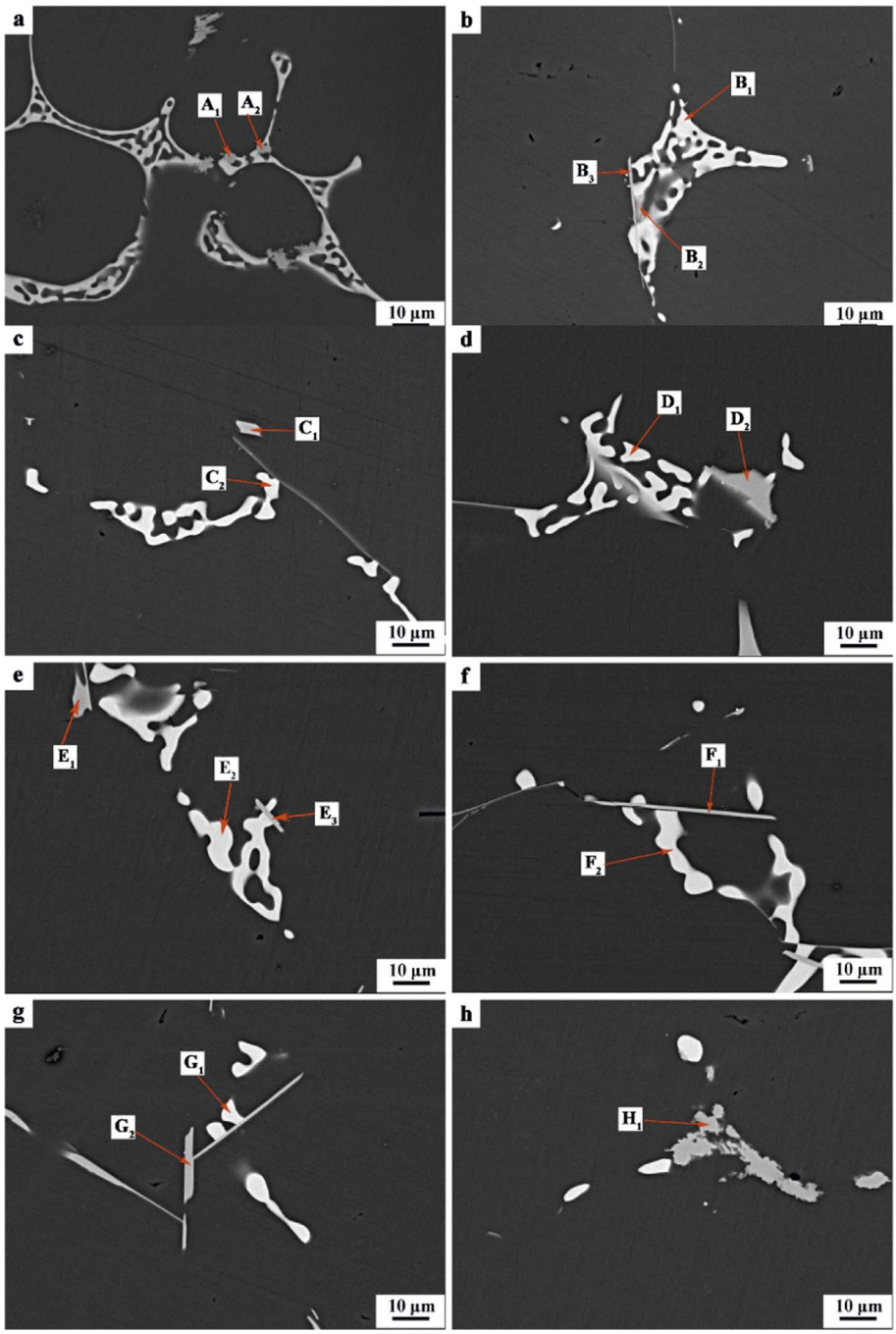

Fig. 6. SEM images of second phases in 7A56 alloy homogenized at 470 °C:
(a) 1 h; (b) 2 h; (c) 6 h; (d) 12 h; (e) 24 h; (f) 36 h; (g) 48 h; (h) 72 h.

discontinuous at some area and eutectic structure was devastated to some extent. As the homogenization treatment continued, the dissolution of AlZnMgCu phase proceeded and gradually reached to a relative balance at about 24 h. During the period of 24 h to 72 h, the deduction of AlZnMgCu phase was not observable due to the low total amount.

To show the amount vibration of AlZnMgCu phase during homogenization, IPP (Image-Pro Plus) was adopted to analyze the remained phase with the SEM images. The relative content of AlZnMgCu phase in

Table 5
EDS result of second phases in Fig. 6 (at.%).

Label	Al	Zn	Mg	Cu	Fe	Phase
A_1	72.10	2.03	0.88	16.59	8.40	Al_7Cu_2Fe
A_2	69.29	2.97	2.33	17.50	7.92	Al_7Cu_2Fe
B_1	30.62	21.63	31.23	16.52	–	AlZnMgCu
B_2	65.31	11.50	13.80	9.39	–	AlZnMgCu
B_3	81.73	1.83	2.02	9.89	4.54	Al_7Cu_2Fe
C_1	72.88	1.22	0.70	16.83	8.37	Al_7Cu_2Fe
C_2	29.12	22.67	31.90	16.31	–	AlZnMgCu
D_1	33.78	20.47	30.45	15.30	–	AlZnMgCu
D_2	72.42	1.82	0.61	17.21	7.94	Al_7Cu_2Fe
E_1	73.26	1.43	0.70	16.47	8.14	Al_7Cu_2Fe
E_2	27.62	22.61	32.75	17.03	–	AlZnMgCu
E_3	71.32	2.94	2.60	15.84	7.30	Al_7Cu_2Fe
F_1	79.20	2.03	1.97	11.76	5.03	Al_7Cu_2Fe
F_2	24.69	23.90	33.24	18.17	–	AlZnMgCu
G_1	25.56	23.28	33.45	17.71	–	AlZnMgCu
G_2	72.50	1.43	1.08	16.67	8.32	Al_7Cu_2Fe
H_1	71.28	1.49	0.93	17.66	8.63	Al_7Cu_2Fe

Table 6
Solid diffusion coefficient of Cu, Mg and Zn in Al-matrix during homogenization (m^2/s).

Element	Calculation Eq. [29]	Temperature	
		380 °C(653.13 K)	470 °C(743.13 K)
Cu	$D = 4.8 \times 10^{-5} \exp(-16{,}069/T)$	9.94×10^{-16}	1.95×10^{-14}
Mg	$D = 6.23 \times 10^{-6} \exp(-13{,}831/T)$	3.96×10^{-15}	5.15×10^{-14}
Zn	$D = 2.45 \times 10^{-5} \exp(-14{,}385/T)$	6.66×10^{-15}	9.58×10^{-14}
$D_{Cu}:D_{Mg}:D_{Zn}$		1:4.0:6.7	1:2.6:4.9

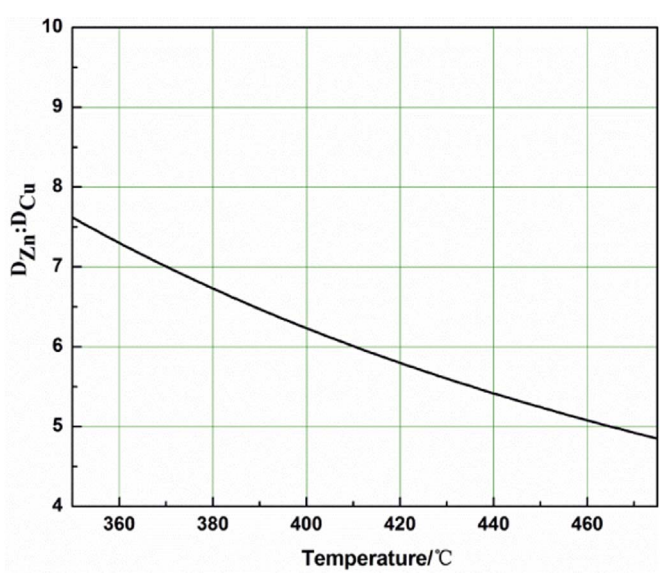

Fig. 7. Diffusion coefficient rate of Zn and Cu in Al-matrix at different temperatures.

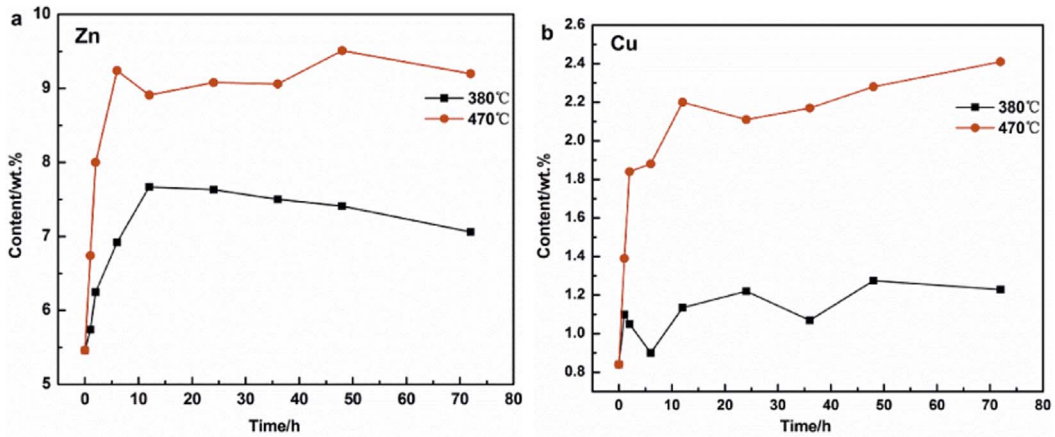

Fig. 8. Content of Zn and Cu in Al-matrix at different temperatures.

original as-cast alloy was set as 100%. As is shown in Fig. 11, the content of AlZnMgCu just fluctuates at a range of 5% at 380 °C. Considering the difference of different samples, it can be merely regarded that the amount did not change overall. By contrast, the relative content of AlZnMgCu phase decreased rapidly within 6 h under 470 °C; then it gradually reaches to a relative balance where the relative amount is lower than 5%, which means more than 95% of the AlZnMgCu phase has been dissolved. After homogenized for 24 h, however, prolonging homogenization time makes no further significant influence on promoting the AlZnMgCu phase to dissolve.

Fig. 12 shows the DSC curves of 7A56 alloy after homogenization treatment. After 380 °C homogenization treatment, there is still an obvious endothermic peaks in every DSC curve and the intensity does not evidently decrease compared with the as-cast state (the 0 h line). After 470 °C homogenization treatment, the endothermic peak gradually decreases and almost vanishes after 24 h homogenization. As the DSC test temperature increases, AlZnMgCu phase will absorb heat for melting [30], the enthalpy of the endothermic peak, to some extent, can reflect the amount of remaining AlZnMgCu after homogenization. The calculated result of enthalpy was shown in Fig. 13. At 380 °C, the enthalpy does not change obviously and keeps a high value even the time was prolonged to 72 h. At 470 °C, it decreases rapidly with the time extending and finally get stabilized at a quite low level (no more than 1 J/g), representing a low remaining of AlZnMgCu phase. The DSC result highly coincides with the IPP analysis and SEM observation (Figs. 11 and 13). Based on all the results above, it can be estimated that more than 95% of AlZnMgCu phase has been dissolved after homogenized 470 °C for 24 h while the total amount of AlZnMgCu phase seldom decreases at 380 °C homogenization treatment.

Fig. 14 shows the electrical conductivity and hardness of 7A56 alloy after homogenization. After 380 °C homogenization treatment, electrical conductivity decreases about 11.8% while the hardness increases about 7.9% after 24 h. Extending the time does not apparently affect the properties. After 470 °C homogenization treatment, the electrical conductivity decreases by 27.5% while the hardness increases by 51.8%. After 24 h, the values of the hardness and electrical conductivity gradually become stabilized. That vibration process highly coincides with the dissolution of AlZnMgCu phase as the latter factor is decisive to the former. With the AlZnMgCu phase dissolved into the matrix, the content of alloying element increases in the interior of Al-matrix. Consequently, the increasing solute atoms can increase the lattice distortion and the dislocation slip resistance. As a result, the hardness of the matrix increased. The solute atoms will become obstacles for electron propagating and thus decrease the electrical conductivity. More AlZnMgCu phase being eliminated, the electrical conductivity and hardness simultaneously change bigger. As the remaining amount of AlZnMgCu phase get stable, the properties will also fall into a balance.

4. Conclusions

1. Temperature can significantly affect the phase transformation of Al_2CuMg from AlZnMgCu in 7A56 alloy. At 380 °C, the lower diffusion coefficient of Cu leads to an easier formation of Al_2CuMg for a majority of Cu atoms were maintained in the pre-existing AlZnMgCu phase. At 470 °C, the diffusion rate of Cu is obviously accelerated and the difference of diffusion coefficient of Cu and Zn is reduced. Consequently, it will result in the direct dissolution of AlZnMgCu instead and not to form the Al_2CuMg phase.
2. Temperature is the main factor on promoting the dissolution of AlZnMgCu phase. Improving temperature is efficient to accelerate the dissolution process as well as raise the total dissolution limits. Homogenized at 380 °C, even the time extended to 72 h, more than 95% AlZnMgCu phase remained and keeps the original eutectic framework. While at 470 °C, the AlZnMgCu phase dissolve rapidly within 6 h; after 24 h, the remaining amount gradually get stable and more than 95% can be dissolved totally.
3. Hardness and electrical conductivity vibration are highly relevant to the dissolution of AlZnMgCu phase. With the time extending and temperature increasing, the hardness increases while the electrical conductivity decreases as more and more AlZnMgCu phase had been dissolved into the Al-matrix. Once the dissolution process reaches into an equilibrium, a state where the total amount of AlZnMgCu

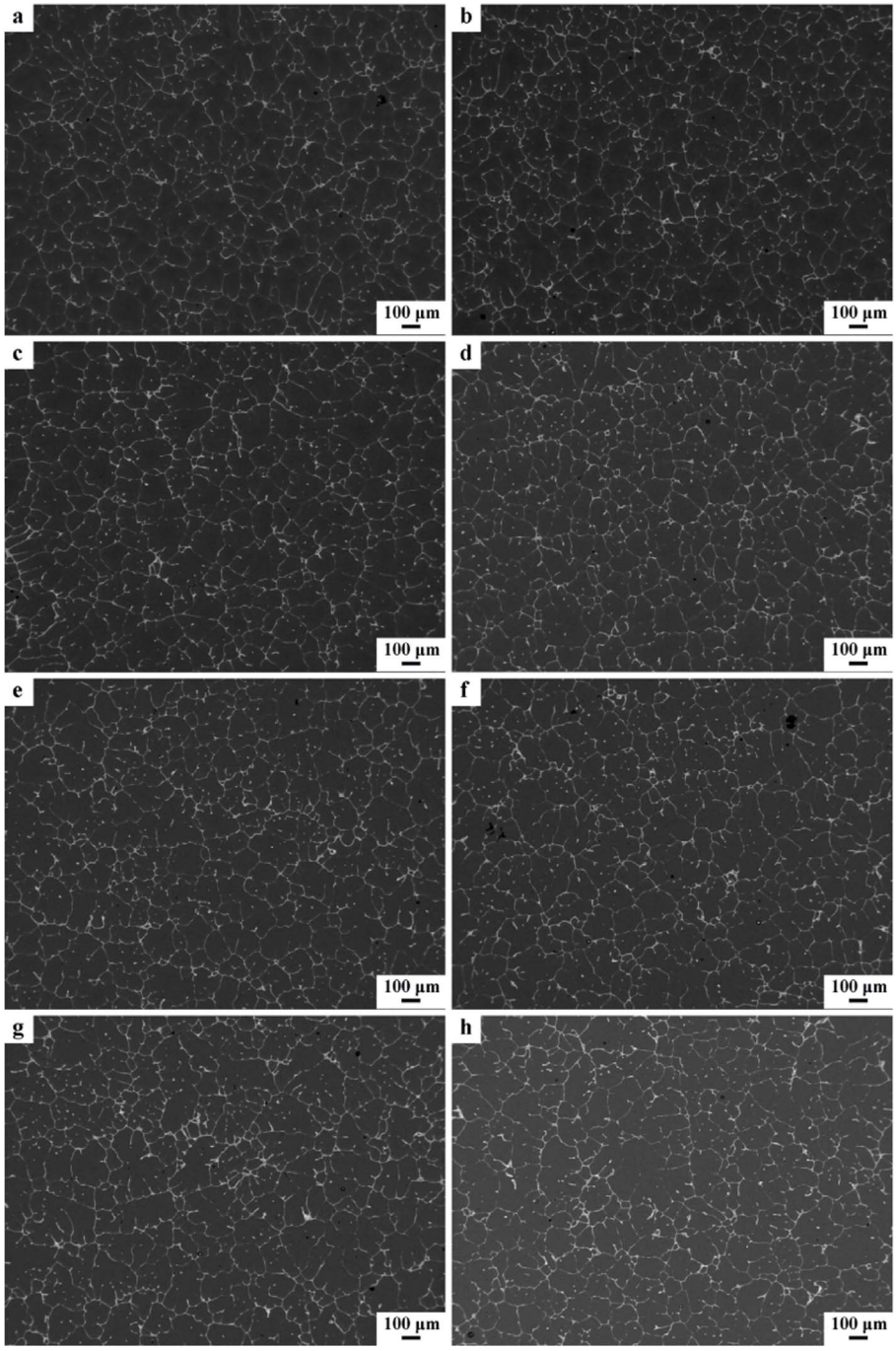

Fig. 9. SEM images of 7A56 alloy homogenized at 380 °C: (a) 1 h; (b) 2 h; (c) 6 h; (d) 12 h; (e) 24 h; (f) 36 h; (g) 48 h; (h) 72 h.

phase is stabilized, the vibration of electrical conductivity and hardness will also turn into stable. The vibration degree highly coincides with the amount of dissolved AlZnMgCu phase.

Acknowledgements

This study is financially supported by the National Key Research and Development Program of China (Grant Nos.2016YFB0300903 and 2016YFB0300803).

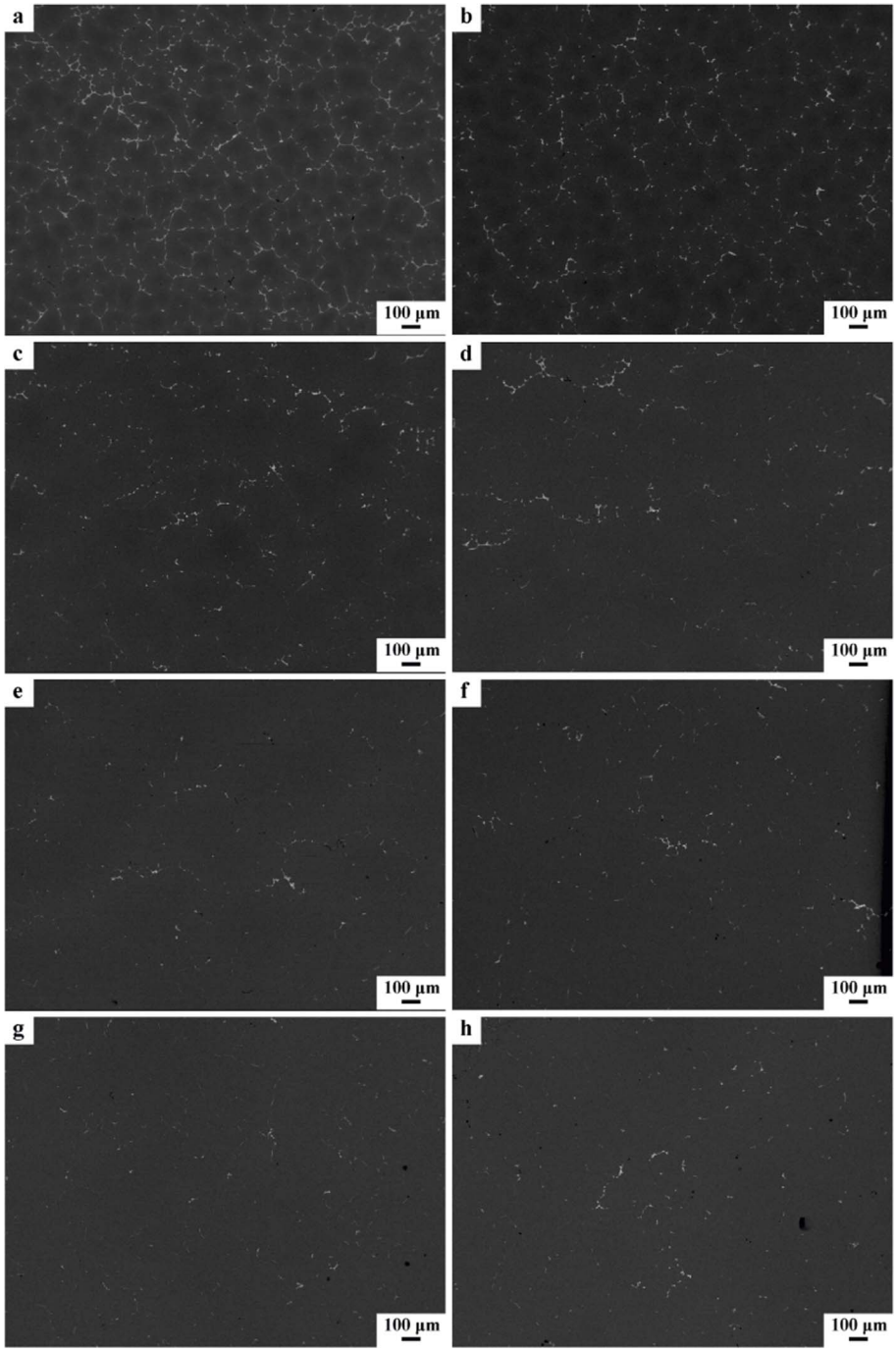

Fig. 10. SEM images of 7A56 alloy at 470 °C:
(a) 1 h; (b) 2 h; (c) 6 h; (d) 12 h; (e) 24 h; (f) 36 h; (g) 48 h; (h) 72 h.

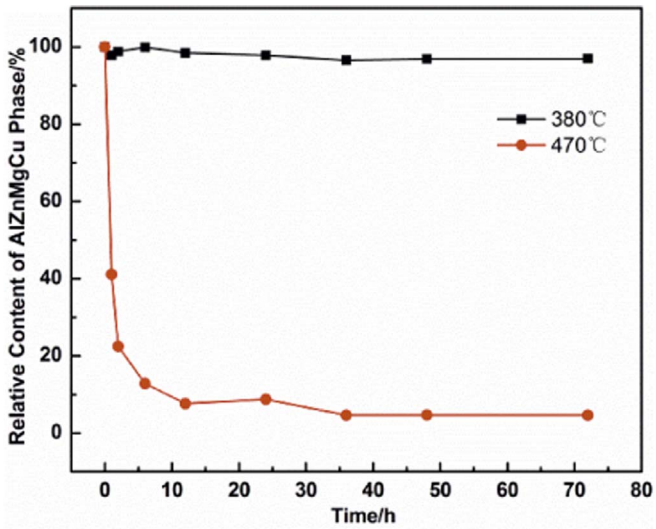

Fig. 11. Relative content of AlZnMgCu phase in homogenized 7A56 alloy.

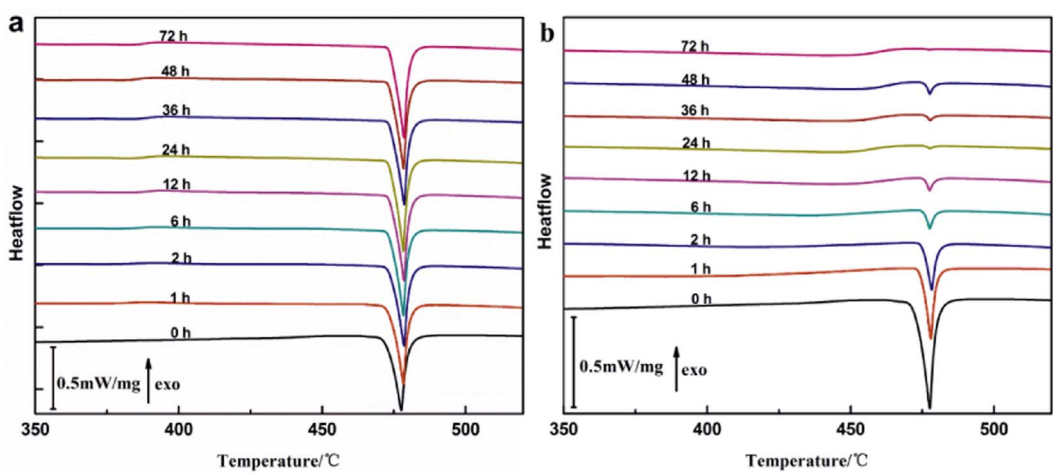

Fig. 12. DSC curves of 7A56 alloy homogenized at: (a) 380 °C; (b) 470 °C.

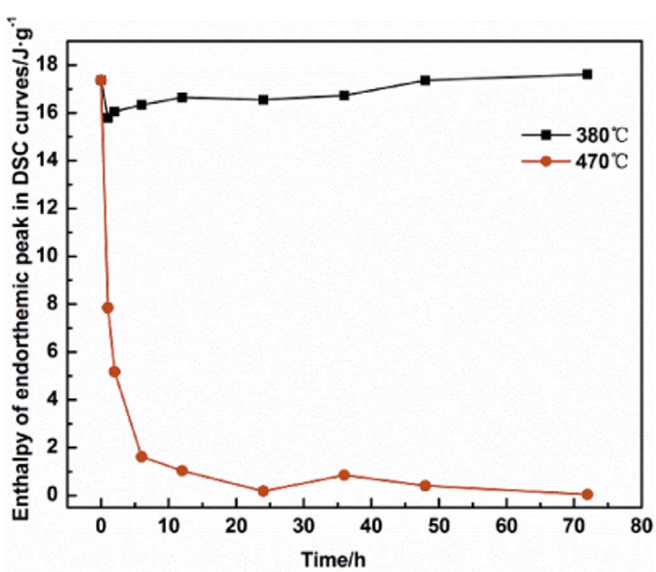

Fig. 13. Enthalpy of endothermic peak in DSC curves.

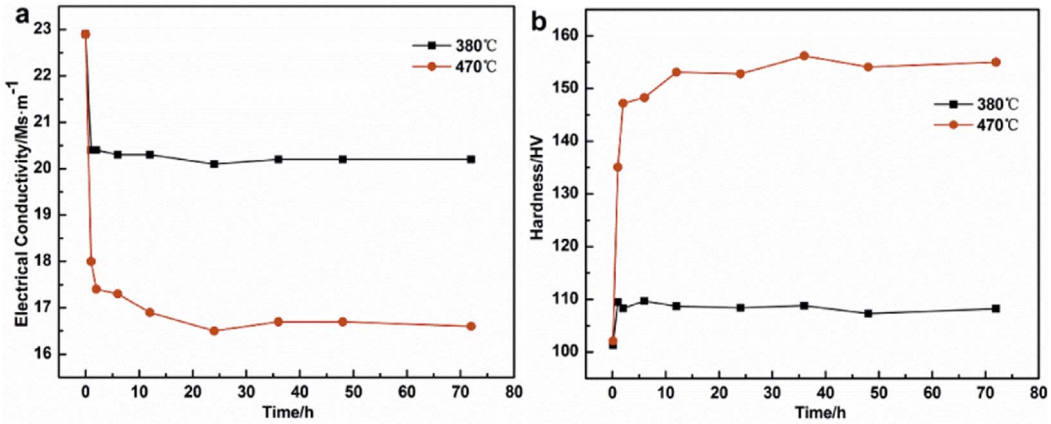

Fig. 14. Electrical conductivity and hardness of homogenized 7A56 alloy:
(a) electrical conductivity; (b) hardness.

References

[1] T. Dursun, C. Soutis, Recent developments in advanced aircraft aluminium alloys, Mater. Des. 56 (2014) 862–871.
[2] A. Sarhan, E. Zalnezhad, M. Hamdi, The influence of higher surface hardness on fretting fatigue life of hard anodized aerospace AL7075-T6 alloy, Mater. Sci. Eng. A 560 (2013) 377–387.
[3] Y. Chen, Y. Yang, Z. Feng, et al., Microstructure, microtexture and precipitation in the ultrafine-grained surface layer of an Al-Zn-Mg-Cu alloy processed by sliding friction treatment, Mater. Charact. 123 (2017) 189–197.
[4] B. Liu, Q. Lei, L. Xie, et al., Microstructure and mechanical properties of high product of strength and elongation Al-Zn-Mg-Cu-Zr alloys fabricated by spray deposition, Mater. Des. 96 (2016) 217–223.
[5] P.K. Rout, M.M. Ghosh, K.S. Ghosh, Microstructural, mechanical and electrochemical behaviour of a 7017 Al-Zn-Mg alloy of different tempers, Mater. Charact. 104 (2015) 49–60.
[6] K. Wen, Y. Fan, G. Wang, et al., Aging behavior and precipitate characterization of a high Zn-containing Al-Zn-Mg-Cu alloy with various tempers, Mater. Des. 101 (2016) 16–23.
[7] Y. Liu, D. Jiang, B. Li, et al., Effect of cooling aging on microstructure and mechanical properties of an Al-Zn-Mg-Cu alloy, Mater. Des. 57 (2014) 79–86.
[8] T. Marland, A. Deschamps, F. Bley, et al., Evolution of precipitate microstructures during the retrogression and re-ageing heat treatment of an Al-Zn-Mg-Cu alloy, Acta Mater. 58 (2010) 4814–4826.
[9] N.J. Henry, G.M. Scamans, Crack propagation during sustained-loaded cracking of Al-Zn-Mg-Cu aluminum alloys exposed to moist air or distilled water, Metall. Mater. Trans. A 42 (2011) 3979–3998.
[10] H. She, D. Shu, W. Chu, et al., Microstructural aspects of second phases in as-cast and homogenized 7055 aluminum alloy with different impurity contents, Metall. Mater. Trans. A 44A (2013) 3504–3510.
[11] R. Ghiaasiaan, X. Zeng, S. Shankar, Controlled Diffusion Solidification (CDS) of Al-Zn-Mg-Cu (7050): microstructure, heat treatment and mechanical properties, Mater. Sci. Eng. A 594 (2014) 260–277.
[12] H. Wang, J. Xu, Y. Kang, et al., Study on inhomogeneous characteristics and optimize homogenization treatment parameter for large size DC ingots of Al-Zn-Mg-Cu alloys, J. Alloys Compd. 585 (2014) 19–24.
[13] Z. Cao, Y. Feng, T. Li, et al., Variation of the microstructure of ingots of DSC cast alloy 7050 during homogenization, Met. Sci. Heat Treat. 52 (2010) 179–182.
[14] N. Li, J. Cui, Microstructural evolution of high strength 7B04 ingot during homogenization, Trans. Nonferrous Metals Soc. China 18 (2008) 769–773.
[15] C. Feng, W. Shou, H. Liu, et al., Microstructure and mechanical properties of high strength Al-Zn-Mg-Cu alloys used for oil drill pipes, Trans. Nonferrous Metals Soc. China 25 (11) (2015) 3515–3522.
[16] P. Bai, X. Hou, X. Zhang, et al., Microstructure and mechanical properties of a large billet of spray formed Al–Zn–Mg–Cu alloy with high Zn content, Mater. Sci. Eng. A 508 (1–2) (2009) 23–27.
[17] J. Li, Z. Peng, C. Li, et al., Mechanical properties, corrosion behaviors and microstructures of 7075 aluminium alloy with various aging treatments, Trans. Nonferrous Metals Soc. China 18 (4) (2008) 755–762.
[18] F. Cong, G. Zhao, F. Jiang, et al., Effect of homogenization treatment on microstructure and mechanical properties of DC cast 7X50 aluminum alloy, Trans. Nonferrous Metals Soc. China 25 (4) (2015) 1027–1034.
[19] N.H. Han, X.M. Zhang, S.D. Liu, et al., Effect of solution treatment on the strength and fracture toughness of aluminum alloy 7050, J. Alloys Compd. 509 (2011) 4138–4145.
[20] G.S. Wang, Z.H. Zhao, Q. Guo, et al., Effect of homogenizing treatment on microstructure and conductivity of 7050 aluminum alloy prepared by low frequency electromagnetic casting, China Foundry. 11 (1) (2014) 39–45.
[21] D.K. Xu, P.A. Rometsch, N. Birbilis, Improved solution treatment for an as-rolled Al–Zn–Mg–Cu alloy. Part II: microstructure and mechanical properties, Mater. Sci. Eng. A 534 (2012) 244–252.
[22] T. Xiao, Y. Deng, L. Ye, et al., Effect of three-stage homogenization on mechanical properties and stress corrosion cracking of Al-Zn-Mg-Zr alloys, Mater. Sci. Eng. A 675 (2016) 280–288.
[23] X. Lv, E. Guo, Z. Li, et al., Research on microstructure in as-cast 7A55 aluminum alloy and its evolution during homogenizaiton, Rare Metals 30 (6) (2011) 664–668.
[24] Y.L. Deng, L. Wan, L.H. Wu, et al., Microstructural evolution of Al-Zn-Mg-Cu alloy during homogenization, J. Mater. Sci. 46 (2011) 811–875.
[25] P. Jia, Y. Cao, Y. Geng, et al., Studies on the microstructures and properties in phase transformation of homogenized 7050 alloy, Mater. Sci. Eng. A 612 (2014) 335–342.
[26] S. Wang, J. Jiang, G. Fan, et al., Microstructure evolution in abrasion-induced surface layer on an Al–Zn–Mg–Cu alloy, Mater. Charact. 98 (2014) 18–25.
[27] W. Li, Q. Pan, Y. Xiao, et al., Microstructural evolution of ultra-high strength Al-Zn-Cu-Mg-Zr alloy containing Sc during homogenization, Trans. Nonferrous Metals Soc. China 21 (10) (2011) 2127–2133.
[28] Y. Liu, D. Jiang, W. Xie, et al., Solidification phases and their evolution during homogenization of a DC cast Al-8.35Zn-2.5Mg-2.25Cu alloy, Mater. Charact. 93 (2014) 173–183.
[29] F. Xie, X. Yan, L. Ding, et al., A study of microstructure and microsegregation of aluminum 7050 alloy, Mater. Sci. Eng. A 355 (1–2) (2003) 144–153.
[30] X.M. Li, M.J. Starink, DSC study on phase transitions and their correlation with properties of overaged Al-Zn-Mg-Cu alloys, J. Mater. Eng. Perform. 21 (6) (2012) 977–984.